

Multimodal Approach for the Diagnosis of Neurodegenerative Disorders Using Augmented Reality

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ABSTRACT

This study introduces an innovative multi-modal strategy for diagnosing neurodegenerative disorders by integrating Augmented/Mixed Reality. Our primary focus involves harnessing AR goggles to capture Parkinson's Disease symptoms, reshaping our comprehension of brain-related disorders. Through meticulous sensor data and technical intricacies like hand movements, gait and gaze analysis, our system offers a transformative approach to patient care. At its core, our work features an autonomous game-like experience — a 30-minute journey encompassing 17 varied tasks seamlessly integrating motor skills, gait and gaze exercises, speech tasks, memory challenges, and cognitive exercises. This gamified platform, coupled with clinical tests, delivers a holistic assessment of patients' physical and cognitive capabilities. By automating tasks and leveraging AR glasses for enhanced patient comfort, our approach not only captivates patients but also emerges as a potent diagnostic tool, heralding an era of interactive neurodiagnostics where innovation converges with patient engagement.

Index Terms: Applied computing—Life and medical sciences—Health informatics—; Mixed Reality—Multimodal signals—Gait analysis—Hand-related analysisSpeech analysisGaze analysisPosture stability;

1 INTRODUCTION

Neurodegenerative diseases comprise a group of congenital or acquired progressive diseases of the nervous system. Parkinson's Disease (PD) is among the most common neurodegenerative conditions, affecting approximately 0.15-0.30% of the general population and around 1.5-2% of individuals aged 70 or above [1]. The high diagnostic error rate (up to 20%) in neurodegenerative disorders, such as Parkinson's disease, stems from clinical symptom similarities and the lack of universally definitive histopathological examinations [2]. While initial pharmacological treatments remain consistent, distinctions in prognosis and non-pharmacological management become significant. Neurodegenerative disorders encompass various aspects, including disturbances in voice and speech, posture, ocular movements, bradykinesia, tremor, and dyskinesias. This study delves into the realm of neurodegenerative disorders, particularly Parkinson's disease, introducing a novel multi-modal strategy through the integration of Augmented Reality (AR)/Mixed Reality (MR). Seamless data fusion from AR/MR glasses allows for an unparalleled exploration of the intricate symptoms associated with Parkinson's disease, offering a new dimension in understanding and diagnosing brain-related disorders. As advancements in medical technology continue to unfold, the significance of early detection and continu-

ous monitoring of Parkinson's disease becomes increasingly crucial. This research endeavors to bridge the gap between technological innovation and healthcare needs, setting the stage for a transformative era in neurodiagnostics and patient care. Importantly, when working with elderly individuals, lightweight and wireless AR glasses can be worn without the need to remove prescription eyeglasses.

2 SYSTEM DESCRIPTION

The objective of this project is to develop an integrated system supported by Augmented/Mixed Reality (AR/MR) technologies for acquiring diagnostic information and monitoring patients with neurodegenerative disorders during treatment based on a set of registered signals (digital biomarkers). The application is created in Unity and runs on Microsoft HoloLens 2.

In neurodegenerative disorders, tracking all symptoms arising from progressive brain neurodegeneration is crucial. Precise, repeatable monitoring of such patients enables swift decision-making regarding their treatment and therapy. This approach aligns with the principles of Medicine 4.0 and Evidence-Based Medicine, emphasizing data-driven, technology-integrated strategies for enhanced healthcare. The project plans to utilize AR/MR glasses to capture signals such as voice and speech, hand movement trajectories, eye movement trajectories, posture stability, and gait. AR glasses feature 5 microphones, 4 visible light cameras, 2 infrared cameras, a depth sensor, and an Inertial Measurement Unit (IMU) sensor (comprising an accelerometer, gyroscope, and magnetometer) for the multimodal recording of biomedical signals.

Hyperkinesias, including tremor, dystonia, and chorea, are symptoms of neurodegenerative diseases. Dystonia is characterized by sustained and repetitive muscle contractions and spasms leading to abnormal, often twisting, postures or repetitive movements. Dystonia may sometimes manifest as gait disturbances, the emergence of specific involuntary movements, and postural instability when combined with stiffness and spasticity in the limbs. Chorea is a movement disorder characterized by involuntary, rapid, and irregular flow from one body part to another and may occur in limbs, face, and trunk. In Parkinson's disease, drug-induced dyskinesias can take the form of both dystonia and chorea.

Utilizing the cameras embedded in AR glasses to track eye movements enables the analysis of their motor functions and reactions to specific tasks displayed to the patient using AR. A tremor is a rhythmic, oscillatory movement of a body part, often involving the hands, arms, head, or other extremities. Tremor may vary in frequency and amplitude and it occurs depending on activation conditions, specifically, resting tremor manifests during complete muscle relaxation and is characteristic of PD, while action tremor appears during muscle activation. However, the presence of tremors alone is insufficient for diagnosis of Parkinson's disease; the key symptom is bradykinesia means slowness of movement. Tremor is a common motor symptom, often the earliest visible one and, with minimal intensity of other symptoms, sometimes the only one contributing to preliminary diagnosis. AR glasses enable hand segmentation

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and monitor their movements. Gait and posture abnormalities are also cardinal signs of PD, but also other neurodegenerative diseases. Using AR glasses IMU enables tracking of parameters indicating stability/abnormalities in posture and gait.

The system addresses the critical need for early detection and continuous monitoring of PD. By automating tasks and leveraging AR technology, clinicians gain valuable time for in-depth analysis, ultimately improving the efficiency of the diagnostic process. Early identification empowers healthcare professionals to implement timely interventions and personalized care strategies, positively impacting the overall well-being of individuals affected by Parkinson's disease.

A fundamental aspect of our system is the introduction of an autonomous immersive experience, engaging patients in a 30-minute journey comprising 17 diverse tasks:

- 5 speech tasks: image description, story repetition, sustained vowels (“a”, “e”, “i”, “o”, “u”) phonation, word repetition (“pataka”, “petaka”, “pakata”), answering the question,
- 4 eye movement tasks: registration of fast movements, including reflex saccades, antisaccades and memory-guided saccades as well as, registration of slow eye movements such as gaze fixation and smooth pursuit.
- 6 hand hand-related tasks: 3 tests to assess postural, kinetic and rest tremor, respectively: (1) extend arms in front and hold for 2 minutes, (2) keep arms in front and alternately touch the nose with each hand; while one hand touches the nose, the other remains extended in front of the patient – perform the exercise for 2 minutes, (3) place your hands on your knees, and gaze at them for 2 minutes; and 3 tests to assess hand bradykinesia (4) raise hands to eye level and demonstrate as if you were screwing in light bulbs (pronation-supination movements), (5) extend the hands in front of you, then open and close your fists (hand movements) (6) tap the index finger on the thumb (finger tapping),
- posture stability and gait assessment: “up and go test”, “10-meter walking test”.

Tasks range from motor skill assessments and gait exercises to speech, memory, and cognitive challenges. This game-like environment captures subtle nuances in patient performance and facilitates a comprehensive evaluation, incorporating established clinical tests. Each task is accompanied by detailed descriptions and motivational messages. The application enhances the user experience by incorporating the calm and informative voice of a lecturer. Additionally, visualizations of the tasks are presented in various formats, including text, audio, and movies, ensuring a holistic understanding of the performed activities. The patient might stop, restart, go back, or next with the voice commands. This multimedia support aims to enrich the user's engagement and overall experience within the application. Figure 1 illustrates the patient's perspective of the displayed instructions for hand movements, while Figure 2 portrays the patient engaging with AR/MR glasses while performing tasks.

3 MEDICAL ASSESSMENT

By automating tasks and leveraging AR technology, clinicians gain valuable time for in-depth analysis, improving diagnostic efficiency. Early identification empowers healthcare professionals to implement timely interventions and personalized care strategies, with medical assessments correlating results from signals with the health state of individuals affected by Parkinson's Disease. This immersive experience includes comprehensive neurological tests such as MDS-UPDRS, Berg Balance Scale, Tinetti's Balance and Gait Evaluation, Timed Up and Go (TUG), Schwab and England Scale, and the Motor Speech Disorder Severity Rating Scale performed by the clinician.



Figure 1: Patient's view of displayed instructions, hand movement task: extend arms in front and hold for 2 minutes.



Figure 2: 68-year-old patient using the AR/MR glasses, performing tasks with hand movement.

4 SUMMARY

By automating tasks, clinicians gain valuable time to focus on analyzing patient performance rather than administering tests. Our observations reveal that patients, immersed in the AR environment, forget the typical anxieties related to tests. This newfound comfort translates into increased eagerness to communicate, enhanced motivation, and an overall more impressive performance. The fusion of 17 diverse tasks not only ensures patient engagement but also serves as a powerful diagnostic tool, providing clinicians with a comprehensive understanding of various neurological aspects. In neurodegenerative diseases, it is crucial to monitor all symptoms arising from progressive brain neurodegeneration. This aligns with the precision medicine model, aiming to tailor an individualized diagnostic and treatment plan for each patient. Thorough monitoring of such patients enables swift decision-making regarding their treatment and therapy. The system represents an innovative approach to diagnosing and monitoring neurodegenerative disorders, focusing specifically on Parkinson's Disease. Augmented/Mixed Reality serves as the core technology, integrating AR goggles with precise sensors to create a comprehensive platform for capturing and interpreting a range of disease-related symptoms.

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